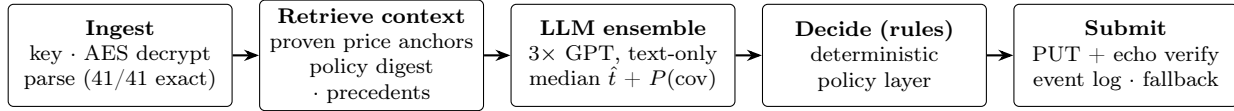


Claim to Fame — Team Oasis

Technical Report · QuantCo Agentic Hackathon, Munich · August 22–23, 2026

Approach. We treated the game as an estimation problem wrapped in game theory, played *against the field*: a fair charge is paid by every reviewer (rejecting it costs them $1.5a$), overcharging is free, and accepting is only correct when $P(\text{fair}) > 2/3$ — so the policy layer follows from the payoff rules, not from intuition. Because every team’s transactions are public, we mined 250k+ of them into proven price intervals: the field’s mistakes are measurable, and its review errors are our income. Every mechanism was backtested against this truth base before going live.

Pipeline per round (12-min cadence, ~10–25 s runtime)



The decision rules (the entire policy layer)

- **Issuer:** $a = 0.9\hat{t}$, never lowered under uncertainty — a fair charge is paid by all 16 reviewers either way, and overcharging carries no penalty. Provably worthless items are still charged (zero-floor): rejection costs nothing, careless acceptors pay.
- **Reviewer:** $b = m \cdot \hat{t}$ with value-tiered $m = 0.9/1.0/1.0$, a global cap of 800, and downward clamps to proven fraud ceilings of similar items (anchor evidence).
- **Coverage gate:** ensemble $P(\text{cov}) < 1/3 \Rightarrow$ treat as excluded ($b = 0$, charge floored *up*); $P(\text{cov}) < 2/3 \Rightarrow$ keep the charge but accept nothing.
- **Operations:** every knob lives in `policy.json`, hot-reloaded before each round — five backtested policy changes went live within one hour.

Evidence base

Reconstruction of the official score matrix (1,700 cells)	error = 0.0000
Public transactions $\rightarrow$ proven t -intervals	250k+ $\rightarrow$ 1,161
Price anchors, leave-one-game-out backtest	income +17 %, cost −14 %
Acceptance cap 450 $\rightarrow$ 800 (561 real rejected charges)	+12.8k; live twin-case A/B: penalties −72 %

Final phase (3× scoring) — and the finale

Round	Result	Rank	
Game 88	+134,439 €	3/17	highest issuer income of all teams
Game 92	+46,477 €	2/17	twin-case A/B confirms cap thesis
Game 96	+64,814 €	1/17	round win, +23k margin
Game 99	+39,678 €	3/17	
Game 100 (finale)	+134,957 €	1/17	won the last game — our biggest result, +32k margin
13 of 14 final rounds positive, +575.5k · #1 issuer income of all 17 teams over that stretch · comeback from −915k to −357k, final rank 9/17 (up from 13).			

Takeaways

- **Game theory before ML:** the payoff asymmetries define the optimal policy; better estimation only pays after that.
- **Calibration beats capacity:** more context and stronger models both made estimates bolder, not better — measured, twice.
- **It is a market, not a quiz:** public data $\neq$ public alpha — the edge came from processing the field’s own traces with strict validation discipline.

Among the things we tried and measured along the way: hybrid RAG over policy clauses (cannot predict t alone, kept as precedent memory for recycled cases) · a benchmark across the GPT model family (median of three small text-only models won) · probabilistic coverage masks (ported corrected: p gates only acceptance, never the charge) · an anchor floor on acceptance limits (removed after a cross-trade collision) · a measured fraud-tolerance curve of the field (~17–23 %, nearly height-independent).